Imad EL BADISY

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PROFILE

Research Engineer in Biostatistics and Machine Learning, PhD, working at the interface of computational biostatistics and machine learning. Moroccan and French national, husband and father. I hold a PhD from Aix-Marseille University (SESSTIM laboratory) on missing data and machine learning imputation methods for survival analysis. My work focuses on building methodological tools for health data science supporting biomedical research projects. I develop open-source R packages for survival analysis, missing data, and interpretable machine learning, and contribute as a consultant and trainer in biostatistics and health data science.

EXPERIENCE

Mohammed VI Center for Research and Innovation (CM6RI) — Rabat, Morocco
Head of AI and Data Science Service — Nov. 2021 – Present

- Leading the AI and Data Science service
- Conducting methodological and applied research in health (oncology, fertility, other biomedical domains)

The GRAPH Network — Switzerland / Remote
Data Science Education Officer — Sep. 2023 – Dec. 2023

- Designed and delivered training programs in data science for global health
- Developed educational material and course content for data science learners

Institut Mines-Télécom, IMT Atlantique — Brest, France
Research Engineer in Applied Statistics — Sep. 2019 – Sep. 2021

- Conducted statistical analyses of national health data (SNDS-based analyses)
- Implemented automated and reproducible data workflows

INSERM — Clermont-Ferrand, France
Biostatistician — Jan. 2019 – Jun. 2019

- Conducted cost-effectiveness evaluation of a clinical protocol on ketamine for chronic pain

Université de Sherbrooke — Sherbrooke, Quebec, Canada
Health Economist Intern — Feb. 2017 – Sep. 2017

- Conducted a systematic review and meta-analysis on health utilities in cancer patients

EDUCATION

Aix-Marseille University, SESSTIM — Marseille, France
PhD in Biostatistics — 2021 – 2025

Thesis: Missing Data and Machine Learning Methods for Survival Analysis · Supervisor: Pr Roch Giorgi

Aix-Marseille University, SESSTIM — Marseille, France
Master 2 in Quantitative Methods for Health Research — 2018 – 2019

Université Clermont Auvergne — Clermont-Ferrand, France
Master 1 in Applied Mathematics and Statistics — 2017 – 2018

Université Clermont Auvergne, CERDI — Clermont-Ferrand, France
Master 2 in Health Economics — 2017

SOFTWARE (R PACKAGES)

survalis — Interpretable Survival Machine Learning Framework — CRAN · v0.7.1

- 19 Survival learners, 7 evaluation metrics, 8 interpretability methods · Maintainer

funcml — Functional Machine Learning Framework CRAN

- Unified framework for functional machine learning in R · Maintainer

survdnn — Deep Neural Networks for Survival Analysis (R torch) CRAN

- Cox, AFT, and discrete-time neural survival models · Maintainer

tvrmst — Time-Varying RMST from Survival Matrices CRAN

- Dynamic RMST summaries for individualized survival curves · Maintainer

unsurv — Unsupervised Clustering of Individualized Survival Curves CRAN

- Phenotyping patients from predicted survival curves · Maintainer

mcstatsim — Monte Carlo Statistical Simulation Tools CRAN

- Functional approach to monte carlo simulation studies in R · Maintainer

missCforest — Ensemble Conditional Trees for Missing Data Imputation CRAN

- Missing data imputation using conditional forest ensembles · Maintainer

PROJECTS

ML4Health — Machine Learning for Health Training 2026

- Training program focused on applied machine learning for health data analysis

AMRPRED — Antimicrobial Resistance Prediction with Machine Learning 2026

- Predictive modeling of antimicrobial resistance using clinical and microbiological data

MELPHY — Melanoma Phenotyping under Immunotherapy and Systemic Treatment 2025 – 2026

- Data-driven phenotyping of melanoma patients treated with immunotherapy and systemic therapies

AIDEM — Artificial Intelligence for Dementia Prognosis and Risk Stratification 2026 – 2027

- AI-based prognostic modeling and risk stratification in dementia using curated clinical data

CLAVUS — Predicting the Clinical Resolution of Variants of Uncertain Significance Using Machine Learning 2026 – 2027

- Machine learning approach for predicting clinical resolution of variants of uncertain significance using curated genomic data

SURVDNN — Survival Deep Neural Networks 2025 – 2026

- Deep learning framework for survival analysis using neural networks

OMOUMATI — Risk Prediction of Pregnancy Complications 2025 – 2026

- Prediction of pregnancy complications to support maternal and infant health outcomes

SELECTED PUBLICATIONS

El Badisy I., Assarag B., Belrhiti Z. (2026). Interpretable clinical decision support in high-risk pregnancy: A scoping review. *BMC Pregnancy and Childbirth.*

Kadi C., Ahmadi N., Houdou A., **El Badisy I.**, et al. (2025). Differentiating latent from active tuberculosis through activation phenotypes and chemokine markers. *Biomarker Insights*, 20, 11772719241312776.

El Badisy I., Graffeo N., Khalis M., Giorgi R. (2024). Multi-metric comparison of ML imputation methods: application to breast cancer survival. *BMC Medical Research Methodology*, 24(1):191.

El Badisy I., BenBrahim Z., Khalis M., et al. (2024). Risk factors for colorectal cancer survival in Morocco: interpretable ML approach. *Scientific Reports*, 14(1):3556.

Houdou A., **El Badisy I.**, Khomsi K., et al. (2024). Interpretable ML for forecasting air pollution: a systematic review. *Aerosol and Air Quality Research*, 24(1):230151.

Zbiri S., Belghiti Alaoui A., **El Badisy I.**, et al. (2024). Private hospitals in LMICs: a typology using cluster methods — Morocco. *BMC Health Services Research*, 24(1):1231.

Blom I.M., Eissa M., Mattijsen J.C., **El Badisy I.**, et al. (2023). Greenhouse gas mitigation interventions for health-care systems. *Bulletin of the World Health Organization*, 102(3):159.

Khalis M., Toure A.B., **El Badisy I.**, et al. (2022). Meteorological parameters and COVID-19 in Casablanca, Morocco. *Int. J. Environmental Research and Public Health*, 19(9):4989.

TEACHING

AI for Public Health — Aix-Marseille University, SESSTIM 2022 – Present

- Machine learning methods and NLP for public health · Master's students

Introduction to Biostatistics — University Mohammed VI of Health Sciences (UM6SS) 2021 – Present

- Descriptive and inferential statistics, statistical reasoning for health sciences · 100+ hours/year

Quantitative Epidemiology and Machine Learning — Institut Mines-Télécom, IMT Atlantique 2020 – Present

- ML fundamentals, quantitative epidemiology, supervised and unsupervised learning · 21+ hours/year

Outside working hours: {**Methods in Health Data Science with R} — an independent online course pathway covering applied biostatistics, survival analysis, machine learning, and meta-analysis with R.**

SELECTED TALKS

Comparison of missing data imputation methods in Cox regression with time-varying effects **2023**

***ISCB 2023, Rome* · El Badisy I. et al.**

Missing data imputation using a meta-algorithm metaCART: simulation under MCAR **2023**

***EpiClin 2023, Nancy* · El Badisy I. et al.**

SKILLS

Methods: Biostatistics · Survival analysis · Missing data · Meta-analysis · Interpretable machine learning

Tools: R package development · Python · Quarto/LaTeX · Git/GitHub · Slurm/HPC

Languages: Arabic (native) · French (fluent) · English (fluent)